

Exercise 11

Advanced SQL 2: Division

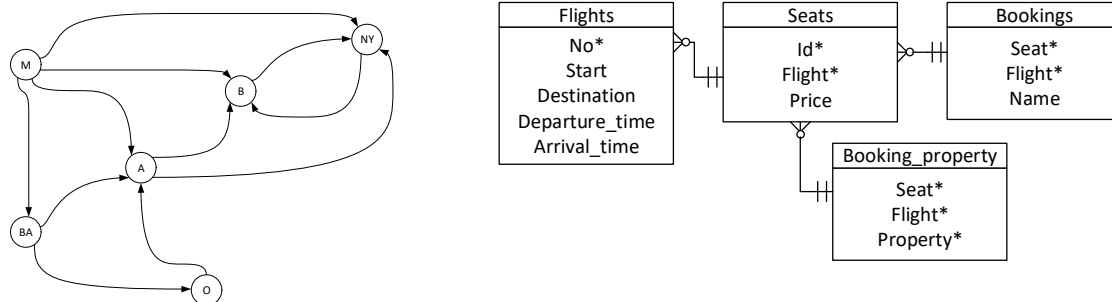
For this exercise sheet, please use the prepared SQL scripts containing the BikeStore data from the lecture DB-Chapter07-KH-Advanced SQL BikeStore Sample DB.zip

Task 1: Division*

- You need to know for which store you need to re-order products. Since the existence of a stock item does not help you (since it could be of quantity 0) and products could not have an entry in stock (Hint: you can use just the inner part of division to solve that problem).
Use a set operation to generate a list of products that are not in stock for every store.
- Now you want to know which products can be bought in every store.
Use division to generate a list of products that are in stock for every store.

Task 2: Division for properties on recursive data

In this task we will combine the topics of recursion and division in order to carry out booking queries in a flight booking system. We are looking for a flight connection from Munich to New York, where there are still second class seats available next to a window and with premium TV. The following flight connections are possible between the cities of Munich, Berlin, Amsterdam, New York, Buenos Aires and Ottawa:



We begin with an Overview of all possible connections from Munich and then expand to an Overview of the seats still available with the required properties. The seats relation contains the seats that are available on a flight at a specific price. In the bookingProperty relation you can find the properties of the respective seats and in the bookings relation you can find the seats that have already been booked.

- First use a recursive query to create an overview of all possible connections from Munich. In order to avoid creating an infinite recursion, you must remember the path taken and react accordingly when you reach a destination that has already been flown on a route.
- Now expand the previously developed query to include the travelling time as well as the departure and arrival times.
- Create an overview of the seats that are still available in general.
- Use division to restrict the previously created view to only those seats that are in second class, next to a window and offer premium TV.